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Optimizing Claim Assessment Processes in Property Insurance: A Case Study

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Abstract

Predicting the property insurance claim has become very important due to the increasing demand for property and the fluctuating number of monthly claims. However, current prediction methods suffer from limited feature important information and imbalanced claim data. This research introduces a machine learning approach by considering demographic factors of insured objects and marketing strategies to predict claim trends. The study also tackles data imbalance using the SMOTE technique. Results reveal that random forest outperforms logistic regression, achieving a recall value of 96%. The research findings enable insurance companies to better analyze insured risk profiles and manage the number of claims.

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1. Introduction

Property insurance protects to individuals and businesses against financial losses arising from property damage or loss due to various events such as fires, natural disasters, or crime. The property insurance market in Indonesia offers promising business opportunities, particularly with the increase in construction lending in 2022, amounting to IDR 692 trillion or a growth of 1.7%, resulting in a 1.58% increase in property demand from the previous year [1]. This presents an opportunity for property insurance companies to expand their market and increase premium revenue.

Despite the growth in insurance premiums, the value of property insurance claims has also increased, reaching IDR 3.99 trillion in Q2/2022, a 42% increase compared to the same period last year [2]. This indicates the need for insurance companies to assess the insured's risk profile carefully during the insurance closing process to reduce the potential for claims. However, predicting when property insurance claims will occur remains a challenge for insurance companies due to limited information related to feature importance and an imbalanced number of claim data compared to policy data.

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For PT XYZ, the number of property insurance claims from 2017-2021 has experienced significant fluctuations. This indicates a need for the company to analyze the potential for property insurance claims from the beginning of the insurance closing process. However, the lack of specific tools to predict claims, limited information related to feature importance, and the imbalanced number of claim data compared to policy data, have made the claims analysis process inaccurate. Therefore, PT XYZ must develop a data analysis system that can identify the variables that affect claim occurrence and classify the insured's risk profile as a consideration in the insurance closing process. With this system, PT XYZ can reduce the potential for claims or manage claim risks, increase customer satisfaction, and reduce claim costs.

The process of risk classification and claim prediction can be done using proven machine learning methods, such as logistic regression analysis and random forest. Machine learning performance is good in various fields such as aviation [3], environment [4-5], and insurance [6]. Logistic regression is a statistical method for modeling categorical response variables with one or more predictor variables [7]. Binary logistic regression analyzes the relationship between the response variable and one or more predictors, with binary qualitative data indicating the presence (1) or absence (0) of a characteristic [8]. Random forest, derived from decision trees, utilizes a voting mechanism for improved classification and regression accuracy [9]. In a study on predicting claims in export credit finance, random forest achieved the highest accuracy among other methods [10]. The feature importance process also needs to be carried out to obtain the ranking of the influence of each variable that affects the model being created. Feature importance calculates the score of features influence on prediction models [11]. It aids in interpreting the impact of each feature [12]. In a study on XGBoost and Feature Importance for Forest and Land Fire Classification, it assists in selecting optimal features [13].

In this research, we propose to compare two machine learning methods and combine them with SMOTE (Synthetic Minority Oversampling Technique) to address the issue of imbalanced class data. SMOTE overcomes imbalanced data by randomly increasing observations of minority classes [14]. It balances the minority class with synthetic data, generated to match the majority class. The number of neighbors is determined for implementation ease. Proximity for numerical data is measured using Euclidean distance, while categorical data utilizes the Value Difference Metric (VDM) [15]. In a study of the effects of data balancing approaches, the model that performed best after the log transformation of the dataset was Logistic Regression paired with the oversampling algorithms SMOTE or ADASYN. [16]. This contributes to better prediction accuracy than conventional estimates, providing new insights for property insurance companies in predicting claims in the future, so that they can reduce the number of claims that occur.

2. Dataset and Methodology

This section aims to detail the dataset and outline the methodology employed to address the research objectives.

2.1 Data Overview and Processing

This study considers the dataset obtained from production data and property insurance claims from PT XYZ, collected from April 2017 to April 2022. Before analysis, the data undergoes preprocessing to identify and rectify missing values and transform the data to meet the research requirements. The preprocessing techniques applied include missing value detection and data transformation. Missing value detection is utilized to identify whether the utilized dataset contains complete or incomplete data, which may affect the outcomes of the property insurance claim prediction model analysis. Table 1 presents that there are three variables, namely construction class, premium, and time period, that contain missing values. As the dataset contains missing values, it is necessary to remove such data to prevent any negative influence on the model being developed.

The subsequent step involves the detection of outlier data on the time period and premium variables. The identified outlier data will be eliminated to construct a more precise prediction model, as shown in Table 2 which displays the data before and after the removal of outliers in the term and premium variables.

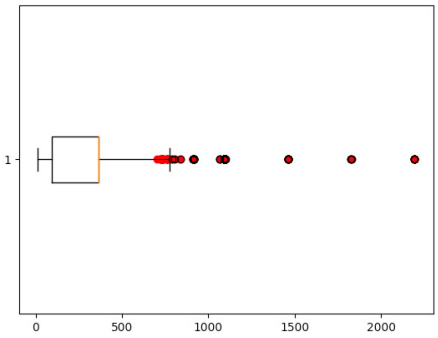
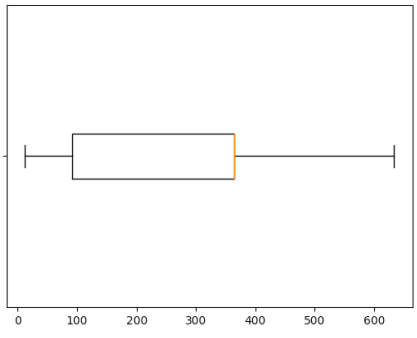
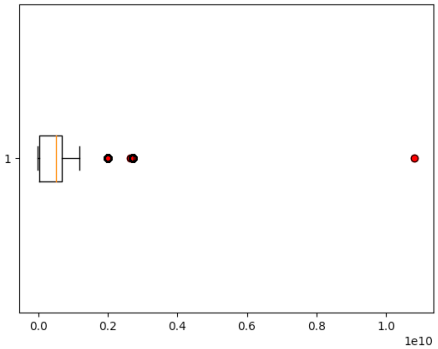
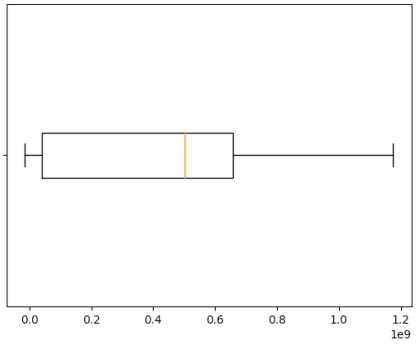
The next step involves data transformation, which is accomplished by applying one-hot encoding to the categorical data, including product type, business source, object location, construction class, flood zone, sprinkler, fire alarm, generator, earthquake zone, and occupation. The process of one-hot encoding transforms categorical data into binary vectors (0/1), representing the presence or absence of specific features in a sample. This step is crucial because

machine learning models can only process numerical variables, and one-hot encoding enhances the model's performance by providing a more accurate numerical representation of the categorical variables.

Table 1. The variables that contain missing values

No	Variables	Total Missing Value	Variable Type	Description
1	Construction Class	216	Ordinal	Building classes categorized by level of risk of damage and repair
2	Occupational	0	Nominal	Usability categories of a building
3	Object Location	0	Nominal	Location of the insured object of coverage
4	Sprinkle	0	Nominal	There is/is no sprinkle
5	Fire Alarm	0	Nominal	There is/is no fire alarm
6	Hydrant	0	Ratio	Number of hydrants available at the insured object
7	Flood Zone	0	Nominal	Is the building located in a flood location
8	Earthquake Zone	0	Nominal	Is the building located in an earthquake location
9	Generator/ No Generator	0	Nominal	Type of lighting in the building (use of generators/no)
10	Premium	29	Ratio	The amount of premium paid by the insured
11	Product Type	0	Nominal	Property insurance product category
12	Agent	0	Nominal	Agent who markets property insurance
13	Time Period	29	Interval	The period the building is insured (in days)
14	Claim/ No Claim	0	Nominal	Whether the insured had a claim / no claim

Table 2. The Changes in The Display of Data Before and After Removing Outliers

Variables	Before	After
Time Period		
Premium		

2.2 Methodology

According to Figure 1, once the problem is identified, a thorough literature review is conducted to explore existing methodologies and approaches that can address the problem effectively. The next step involves preprocessing the available dataset by removing any missing values and outliers. This ensures that the dataset is clean and ready for further analysis and prediction model development. The dataset will be divided into training and testing data before

the prediction model is built, with 70% for training data and 30% for testing data, respectively. There does not seem to be clear guidance on what ratio is best or optimal for a given dataset and depends primarily on the loss function that is used [16-17]. An analysis of the models' performance revealed that the training/testing ratios had a significant impact on the predictive power of machine learning models, with the 70/30 ratio showing the models' best performance [18]. The 70% split of testing data was chosen to reduce the risk of overfitting as the model is too "familiar" with the training data and is unable to generalize well to new data. While 30% becomes the testing data to measure the extent to which the model can generalize and its performance outside the training sample.

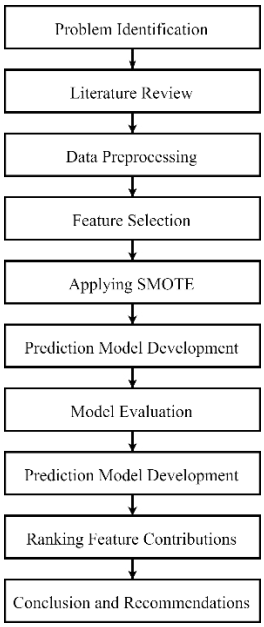


Figure 1. Research flow chart

The classification prediction model is developed using logistic regression and random forest algorithms. The classification model is evaluated using a confusion matrix that has several value parameters, including accuracy, precision, recall, and f-1 score. Accuracy represents the proportion of correctly classified data out of the total amount of data, while precision represents how accurately the positive class (Claim) predicts the positive class (Claim). Recall, also known as sensitivity, indicates how well the model can predict the positive class. In conclusion, the f1-score is a metric that considers both precision and recall. The higher the f1-score, the higher the precision and recall, making it a better measure than accuracy.

To improve the accuracy of the classification model, the SMOTE process is implemented by adding training data to the minor classes to form a balanced dataset. Next, the models built before and after SMOTE are evaluated, and the model with the highest recall value is selected. This model is further broken down to obtain the importance score of each feature. The important features will then be used to provide recommendations during the insurance closing process to prevent insurance claims.

3. Result and Discussion

This chapter presents in detail the results of our research, including significant data and findings, and provides an in-depth analysis of the implications of these results for the research questions posed.

3.1 Logistic Regression and Random Forest Model Before SMOTE

Table 3 demonstrates that the random forest classification model has superior performance compared to the logistic regression model. There are several parameters for comparison of the logistic regression and the random forest including accuracy, precision, recall, and F1-score.

Table 3. The result of the confusion matrix for the built classification model.

Metric	Logistic Regression	Random Forest
Accuracy	1.00	1.00
Precision	0.98	1.00
Recall	0.91	0.92
F1-score	0.94	0.96

3.2 Logistic Regression and Random Forest Model After SMOTE

Figures 2 and 3 show the condition of the number of claims data before and after the SMOTE process is carried out. To address the imbalanced dataset with a large number of claims despite their small percentage, the Synthetic Minority Over-sampling Technique (SMOTE) is considered before performing classifier. This enables the classifier to learn effectively from both the majority and minority classes, resulting in enhanced predictive performance and a more precise analysis of claim trends.

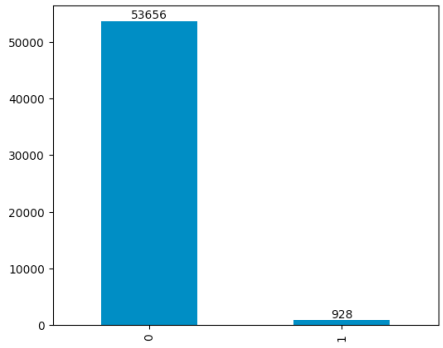


Figure 2. Number of Claims Before SMOTE

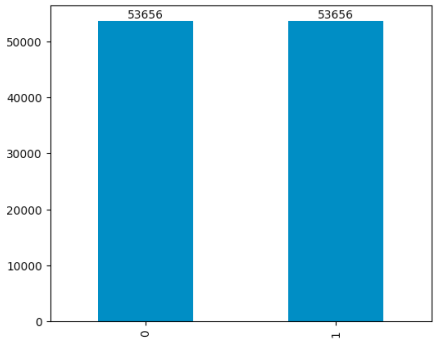


Figure 3. Number of Claims After SMOTE

After performing SMOTE, the classification model was reconstructed using logistic regression and random forest algorithms, resulting in the following confusion matrix.

Table 4. Confusion Matrix Logistic Regression and Random Forest After SMOTE

Metric	Logistic Regression	Random Forest
Accuracy	0.99	1.00
Precision	0.55	0.80
Recall	0.94	0.96
F1-score	0.70	0.87

Table 4 demonstrates that, following the SMOTE process, the random forest classification model still yields better results compared to logistic regression. Consequently, the next step involves comparing the performance of the random forest classification model before and after implementing the SMOTE process.

Table 5. Confusion Matrix Random Forest Before and After SMOTE

Metric	Before SMOTE	After SMOTE
Accuracy	1.00	1.00
Precision	1.00	0.80
Recall	0.92	0.96
F1-score	0.96	0.87

When examining the comparison between the Random Forest metric measurements before and after applying SMOTE in Table 5, it can be observed that the accuracy, precision, and F1-Score values decrease, while the recall value increases. This reduction in accuracy may occur due to the "accuracy paradox" resulting from unbalanced claim data. When the majority class has more data than the minority class, the model may "opt" to always predict the majority class to achieve a high accuracy score, but in reality, it may not be able to accurately identify the minority class. Table

5 demonstrates that the recall value increases after the SMOTE process. The recall or sensitivity value demonstrates the model's ability to minimize false negatives, which are insurance claim predictions that are genuine but are predicted to be invalid by the model. This is critical for PT XYZ to ensure that the company can provide prompt claims services and decrease the risk of losing customers due to the inability to provide appropriate compensation values when claims occur.

3.3 Feature Importance

The table depicted in 6 illustrates the order of feature importance scores for each independent variable obtained from the classification model generated using the Random Forest method after implementing the SMOTE process.

The calculation of feature importance is done with the Random Forest Classifier method which is calculated based on the concept of Gini impurity or mean decrease impurity. Gini impurity measures the impurity or irregularity of a node in the decision tree [19]. Features that consistently contribute to reducing impurity and improving model performance will have a higher importance score, indicating their importance in making predictions.

Table 6 illustrates the importance scores of the 8 features with the highest impact, which account for 81% of the total features. The most influential features include construction classes III and I, marketing business source, flood zones 1 and "not in a flood zone," earthquake zone 4, and time period.

Table 6. Top 8 Feature Importance

	Feature	Importance
0	Construction class (construction class III)	0,18
1	Construction class (construction class I)	0,17
2	Business source (Marketing)	0,11
3	Flood zone (Zone 1)	0,09
4	Flood zone (Not in a flood zone)	0,09
5	Occupation group (Commercial)	0,07
6	Earthquake zone (Zone 4)	0,06
7	Time Period	0,04

Construction classes III and I significantly influence the likelihood of claims because they are closely related to the type of building construction. Buildings in construction class III are more prone to claims because they have more leeway or are not built with non-combustible materials compared to buildings in construction class I, which have walls, floors, structural support components, and roof coverings made entirely of non-combustible materials. The business source also affects the potential for claims to occur because insurance marketing agents tend to sell policies to higher-risk owners, while brokers are trained to select clients with lower risks. Flood zones and earthquake zones are critical features because they are associated with the risk of financial loss in the event of a claim. Areas prone to floods or earthquakes have a higher risk of damage or loss of insured property. The time period is also important to consider in predicting insurance claims because longer periods increase the likelihood of claims and the extent of property damage or loss.

4. Conclusion

The number of property insurance claims is dominated by insured objects located on the island of Java, PSAGBI product type, marketing business source, construction class III, commercial occupation group, not having sprinkle and fire alarms on the insured object, and earthquake zone 4. The prediction model using classification (claim/no claim) is built in 2 stages, namely before SMOTE and after SMOTE. The model was evaluated using a confusion matrix with several parameters namely accuracy, precision, f1-score, and recall. It is found that the best model is a random forest after the SMOTE process is carried out, this is seen from the highest recall value of 96% which states that the model can predict 96% of valid insurance claim data but is predicted to be invalid. The recall value is the main consideration because PT XYZ as an insurance company needs to minimize the risk of false negatives or claims that should be valid but are predicted to be invalid. Based on the results of the feature importance calculation in the previous chapter, it is found that the features that have the highest importance scores are construction class, business source, flood zone, occupation group, earthquake zone, and time period. After obtaining the feature importance value, PT XYZ can prioritize features that have a high level of importance as a consideration when closing insurance. Insured objects that are included in construction class III, marketing business sources, commercial occupation groups, earthquake zone 4,

and long time periods need to be aware of when closing insurance because they have more potential to make insurance claims. For future research, it is recommended to explore more advanced ensemble methods, such as Gradient Boosting Machines (GBM) or XGBoost, to improve the prediction accuracy of property insurance claims [20]. In addition, incorporating external data sources, such as weather indicators, socio-economic and insured profile data, can provide valuable insights into risk factors. Evaluating alternative sampling techniques, such as SMOTE-ENN or ADASYN, can help address the problem of imbalanced data more effectively [21].

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